

Table 4.3: Estimated queue length

Maximum stopping time (minutes)	Multiplier	
	Ma (multiplier for average vehicles)	Mo (multiplier for oversized vehicles)
2	2.4	8
5	6	20
10	12	40
15	18	60
30*	36	120

*A 30 minute stop time is unusual but has been included for some circumstances

For the purpose of estimating the end-of-queue position only, the term 'oversized vehicles' also includes 'heavy vehicles'.

When estimating the queue length, Table 4.3 uses an average vehicle length of 6 m and an average heavy vehicle length of 20 m for a five minute stopping time (based on the five minute traffic count) and then adjusts the multipliers for other stopping time values.

Where very long vehicles are expected (for example outback and mining routes or routes where most vehicles are towing), then to ensure estimated queue lengths are as accurate as possible, an average length per vehicle that matches the actual traffic mix expected at that location should be used.

The estimated queue length must be at least a minimum of one average vehicle plus one heavy vehicle (of a length which may be applicable to the site). Part lengths of vehicles must be rounded up when estimating queue lengths.

To calculate the 'maximum stopping time' value used in Table 4.3 for each approach, the Traffic Management Designer must estimate the likely duration of time that queued traffic will be stopped at a traffic control station. Calculating the maximum stopping time needs to include the total time from when the traffic controller stops traffic, through until the same traffic controller releases traffic for the next cycle from that approach. This would typically include the following considerations:

- the time taken for the traffic queue from one approach to pass the traffic control station and travel the length of the closure
- the time for the queue at the other end of the site to leave that traffic control station and also travel the length of the closure (this may commence at the same time as the other queue or following the completion of the other queues travel if shuttle flow is in operation)
- if traffic queues are held for a period (with no traffic traveling through or past the site), then this hold time will also need to be included